Assignment 1 — Autonomous drone mission (A1.1 – A1.3)

Team: Pšenák, Valko · **Repository:** `LRS_Psenak_Valko`, branches `A-star` (A1.1), `A1_2` (A1.1 + A1.2), `A1_3` (A1.1 + A1.3) · ROS 2 Jazzy, ArduCopter SITL, Gazebo, MAVROS

One ROS 2 package, `lrs_psenak_valko`. Indoors it loads the 3D map of the hangar, plans a collision-free route with A* and flies a mission from a file (A1.1, A1.2). Outdoors it flies, from GPS and compass, a square with the nose on a point of interest and a figure-8 (A1.3). This document covers all parts; the more detailed per-part documents are in `docs/A1_1`, `docs/A1_2` (branch `A1_2`) and `docs/A1_3` (branch `A1_3`).

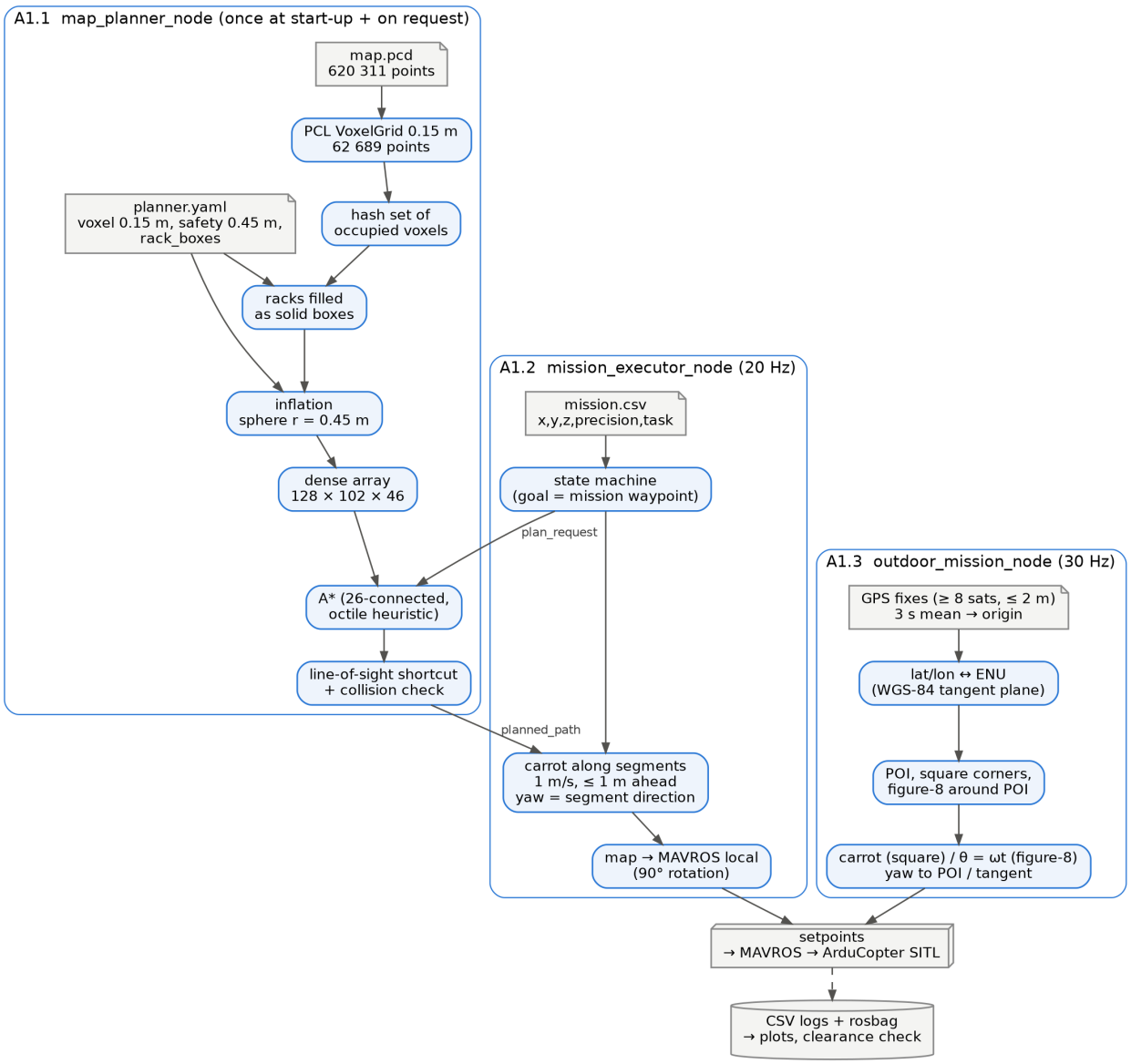
How to run and what is submitted

Part	Branch	Command	Flight record
A1.1 planner + RViz	`A-star` / `A1_2`	`./run_a1_1.sh [--start X Y Z --goal X Y Z] [--safety-radius R]`	numbers in `docs/A1_1/data/`
A1.2 indoor mission	`A1_2`	`./run_a1_2.sh [mission.csv]`	`results/A1_2/safety_0.45_20261018/` (rosbag, CSV, plots)
A1.3 outdoor mission	`A1_3`	`./run_a1_3.sh --record`	`results/A1_3/run_20261017_212049/`

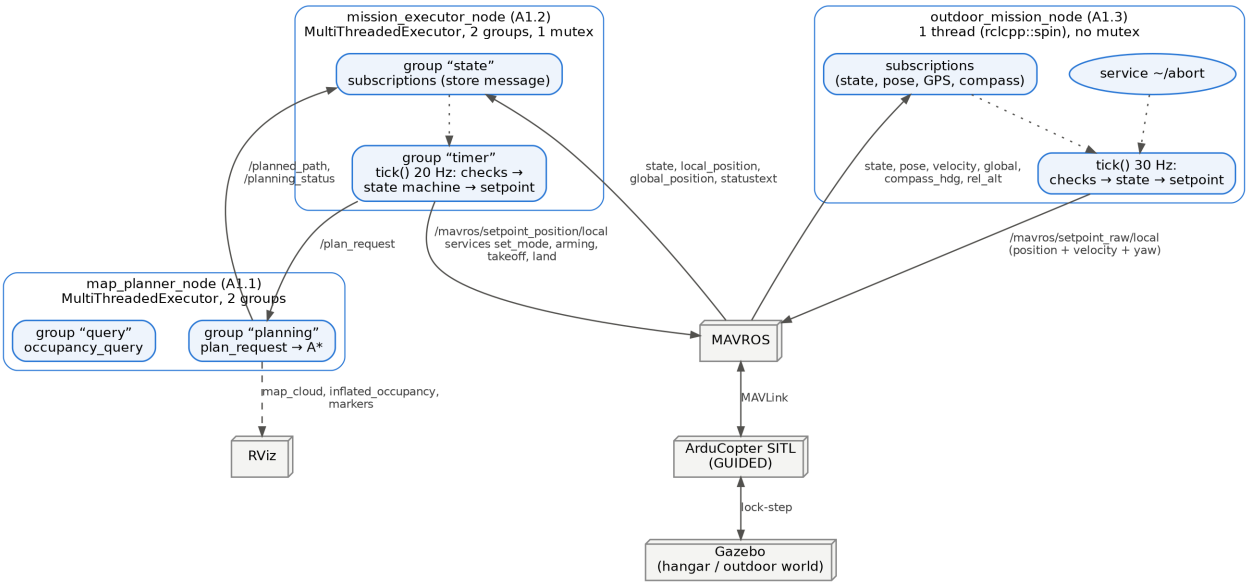
The scripts build the package (`colcon build`) and start Gazebo, SITL and MAVROS in separate terminals. Missions: `src/lrs_psenak_valko/missions/*.csv`. Plots and 3D view of a recorded A1.2 flight: `./show_a1_2.sh`.

Overall solution diagram

Data flow: the map is processed once when the planner starts. The mission holds only goal points; A* computes the route between them and the executor follows it with a "carrot". The outdoor mission takes its origin from GPS once and then flies on the local position.



ROS diagram: the three nodes, their topics and services, and how they are threaded. The planner and the indoor executor use a `MultiThreadedExecutor` with two callback groups (the `mt_executor_demo` pattern): long planning never blocks queries, and a burst of telemetry never delays the control timer. The outdoor node runs in one thread on purpose.



A1.1 — Map and path planning

Map: 0.15 m voxel grid

What: `map.pcd` (620 311 points) is loaded with PCL, downsampled with `VoxelGrid` to 0.15 m (62 689 points) and the occupied voxels are stored in a sparse hash set (key = three 21-bit indices in one `int64`). For A* the occupancy is copied into a dense 128 × 102 × 46 array. Map bounds: x [−0.30, 17.95], y [−1.15, 13.20], z [0, 5.95] m. Any point can be queried through `/occupancy_query`.

Why: planning over raw points would not finish in reasonable time. 0.15 m is a compromise: finer = more voxels for A*, coarser = narrow passages close. The grid is fully 3D (not stacked 2D layers).

Pros / cons: + O(1) query, the sparse set saves memory. – A voxel rounds positions: an obstacle can be up to ~0.075 m closer than the voxel centre suggests.

Racks as solid boxes

What: both racks are filled as solid boxes (`rack_boxes` in `planner.yaml`: centre, size, yaw), 6.50 × 1.05 × 3.76 m centred at (4.32, 6.82) and (4.32, 11.36), from the collision shapes in the Gazebo world. The node refuses to start without them.

Why: the cloud only sees shelf surfaces; between the levels the voxels look free. In the current world file the racks are 3.76 m high (not the 5.59 m quoted in the assignment), so flying **over** a rack (z ≥ ~4.1 m) is valid.

Obstacle inflation: safety_radius 0.45 m

What: every occupied voxel grows into a sphere of radius `safety_radius` in 3D (also above and below); the drone is then a point.

Why 0.45 m: drone radius to the propeller tips 0.33 m (body 0.35 × 0.35 m, propellers at ±0.15 m) + ~0.12 m for position tracking error. **Verified in flight on the same mission:** with 0.30 m the drone came within 0.20 m of the rack edge on the descent from the rack (the propellers would hit it); with 0.45 m

the smallest clearance was 0.37 m.

Pros / cons: + simple and safe also above and below obstacles. – A larger radius closes passages and blocks mission points closer than ~0.45 m to a wall (handled by "snapping" up to 0.75 m).

A* with the octile heuristic

What: A* on the dense array, 26 neighbours, step cost = step length (1, √2, √3 voxels). Heuristic: axis distances sorted d1 ≤ d2 ≤ d3, `h = voxel · [(√3 − √2)·d1 + (√2 − 1)·d2 + d3]`. A diagonal step is allowed only if the whole cube it crosses is free (no corner cutting).

Why: h is the exact shortest-path length in empty space and obstacles can only make the path longer, so it is **admissible** and also **consistent** → the route is optimal on the grid. Euclidean would also be admissible but weaker (more expansions); Manhattan would overestimate on diagonals.

No route: the planner always terminates and says why: `START/GOAL_OUT_OF_BOUNDS`, `START/GOAL_OCCUPIED`, `NO_PATH` (open list exhausted) or `TIMEOUT` (5 s). The report also gives a hint (nearest free point).

Path simplification and collision check

What: line-of-sight shortcutting: from point i jump to the farthest point j reachable by a free segment (sampled every quarter voxel). The node then checks the result again, independently, against the inflated map and never publishes a colliding route.

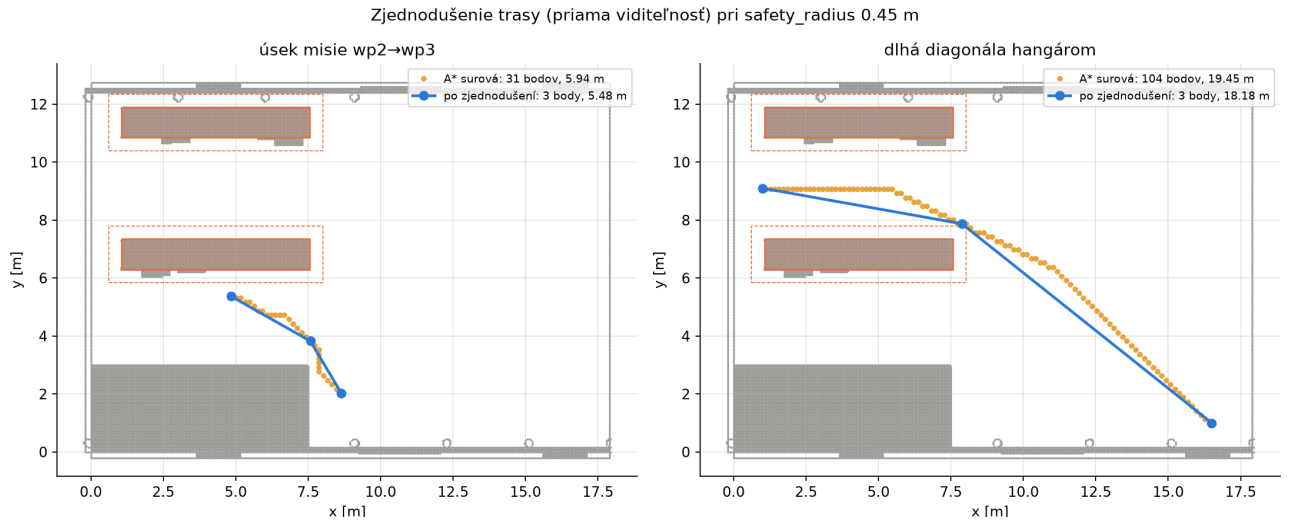
Why: the raw route is a staircase of voxels; in A1.2 every point would mean slowing down. Shortcutting is also exactly the step that can cut a corner, hence the second check.

A1.1 results (no simulator, `tools/a1_1_benchmark.py`)

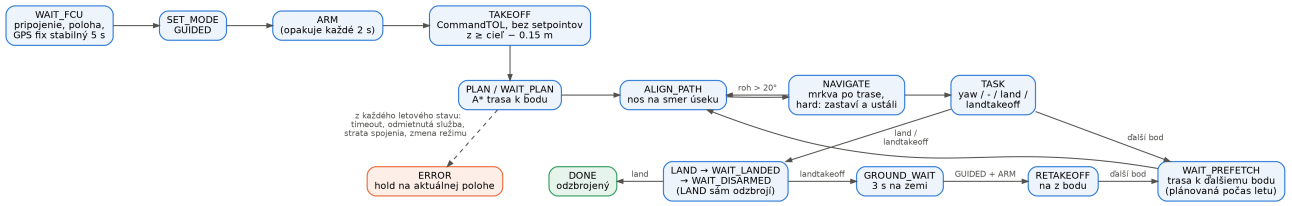
Map: 73 069 occupied voxels with racks, 261 759 after inflation by 0.45 m.

Route (start → goal, m)	A* points → simplified	Length A* → final	Time
around a rack (13, 7, 1.5) → (4.5, 9.1, 1.5)	58 → 2	9.42 → 8.76 m	4.4 ms
over a rack (12, 4, 1) → (4.3, 6.8, 4.8)	53 → 3	10.30 → 9.27 m	46.4 ms
mission leg (8.65, 2.02, 1) → (4.84, 5.37, 2)	31 → 3	5.94 → 5.48 m	14.7 ms
long diagonal (16.5, 1, 1) → (1, 9.1, 3)	104 → 3	19.45 → 18.18 m	8.3 ms
goal in a rack / outside the map	`GOAL_OCCUPIED` / `GOAL_OUT_OF_BOUNDS`	—	0.0 ms
goal in the closed office	`NO_PATH` (331 458 voxels)	—	618 ms

Effect of the radius on the route over the rack: 0.30 m → 9.12 m, 0.45 m → 9.27 m, 0.75 m → 9.55 m (the route climbs higher). Every successful route passed the independent collision check.



A1.2 — Indoor mission



Mission file and conventions

What: the mission path is the ROS parameter `mission_file` (`x,y,z,precision,task`; tasks `-`, `takeoff`, `land`, `landtakeoff`, `yaw<angle>`). A bad line stops the node on the ground with the line number. Rows are goals; A* plans the route between them (no hardcoded points).

Frame	Where	Axes
`map`	mission, planner, plots, Gazebo	x = north, y = west, z up
MAVROS local	pose, setpoints	ENU (x = east), rotated by 90° from `map`
ArduPilot	inside the FCU	NED, ENU ↔ NED is done by MAVROS

`yaw<angle>` is an **absolute** angle in the `map` frame, positive counter-clockwise, 0° = +x. The 90° `map ↔ local` rotation was verified: Gazebo (12.99, 7.00) = MAVROS (–7.00, 12.99).

Start-up: readiness, GUIDED, arming, takeoff

What: we wait for the connection, a local position and a stable GPS fix. Services (set_mode, arming, takeoff, land) are called asynchronously with a request number, and we continue only once `/mavros/state` confirms the real state; no reply within 5 s → ERROR. **No position setpoints are sent during takeoff.**

Why: in GUIDED a position setpoint cancels a running NAV_TAKEOFF, and the drone disarms itself after 10 s on the ground. FCU messages (`PreArm: …`) are copied into our log, so the reason for a refusal is visible.

Carrot following and heading

What: the setpoint (carrot) slides along the straight segment at 1 m/s and is never more than 1 m ahead of the drone. Intermediate route points count as passed within 0.45 m, **without stopping**; only mission waypoints have a precision. The next leg is planned while the current one is flown. The nose points along the segment (`atan2(Δy, Δx)`); at corners sharper than 20° the drone first turns on the spot. The yaw command is kept continuous (unwrapped): from 170° to −170° the drone turns by 20°.

Why: when we sent only the segment end, ArduPilot left the straight line by up to 0.56 m (descending early over a rack). The A* route is collision-free only on the straight line.

Pros / cons: + the drone stays on the route and does not stop at intermediate points. – 1 m/s, short stops at sharp corners.

Hard and soft precision

	Tolerance	Reached when
`hard`	0.15 m	within 0.15 m **and** speed ≤ 0.2 m/s continuously for 1 s (at least 2 distinct velocity messages)
`soft`	0.55 m	within 0.55 m, no stop

Why: 0.15 m is in the recommended 10–20 cm range and is reached reliably in SITL (2.9–9.7 cm). The "stable for 1 s" condition rules out a fly-by that is inside the tolerance for a single sample. Distance from obstacles is ensured by `safety_radius`, not by the tolerance.

Tasks yaw, land and landtakeoff

Yaw: target = the given absolute angle, done when the error is ≤ 5° for 0.5 s; the angle is sent in the position setpoint (latched in GUIDED, it does not expire like a velocity command). **Land:** `cmd/land`; after touchdown we wait for LAND to **disarm the drone by itself**; our own disarm only after 5 s and with retries (ArduPilot refuses it until its land detector confirms touchdown). **Landtakeoff:** 3 s on the ground → GUIDED → ARM → takeoff to the waypoint height.

Threads and errors

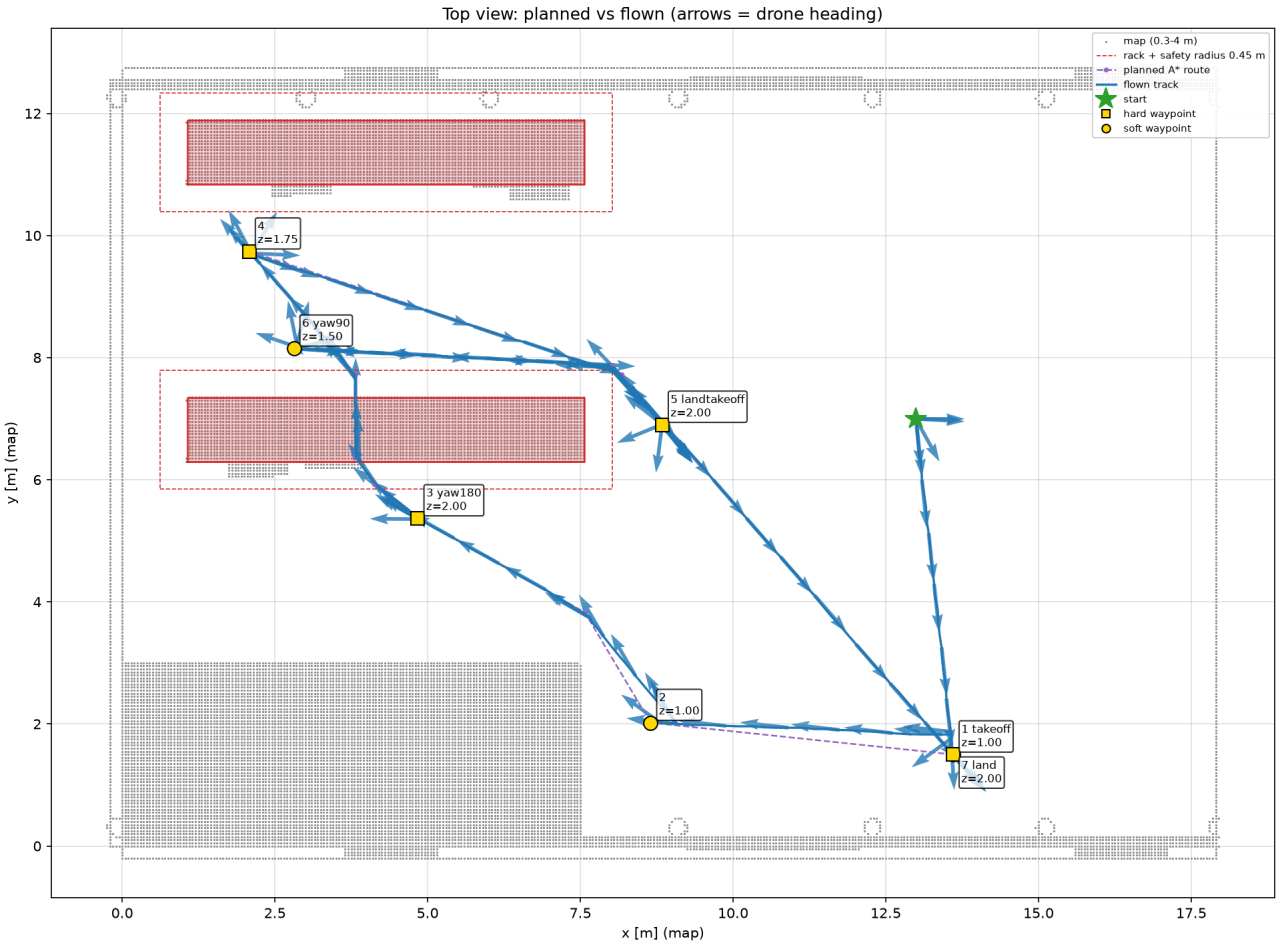
What: `MultiThreadedExecutor`, group "state" (subscriptions only store the message) and group "timer" (`tick()` at 20 Hz: common checks → state machine → setpoint). Shared data is protected by one mutex. ERROR (the drone hovers at its current position) on a state timeout, a rejected service, a position older than 5 s, a lost connection, an unexpected disarm or a mode other than GUIDED (we do not fight the pilot). With a slow simulation the script measures Gazebo's speed and scales the timeouts.

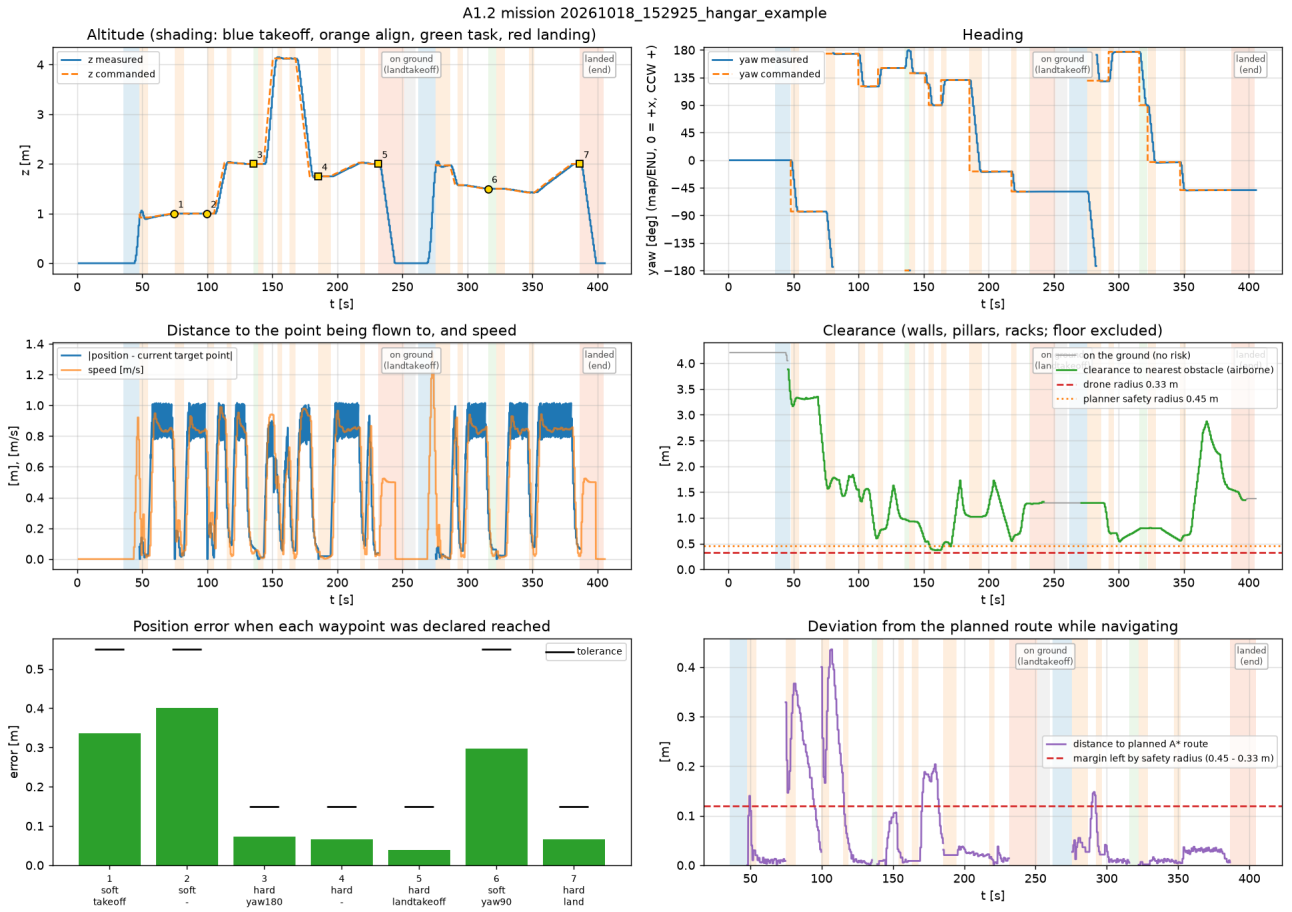
Pros / cons: + control runs exactly 20 times a second. – The mutex is necessary.

A1.2 results (SITL, safety_radius 0.45 m)

Mission	Points	Result	Hard error	Soft error	Min clearance
`hangar_example.csv` (recorded)	7	DONE, disarmed	3.9–7.3 cm	0.30–0.40 m	0.37 m
`test_east_aisle.csv`	8	DONE, disarmed	2.9–7.0 cm	0.07–0.34 m	0.61 m
`test_over_racks.csv`	7	DONE, disarmed	5.0–7.6 cm	0.30–0.35 m	0.38 m

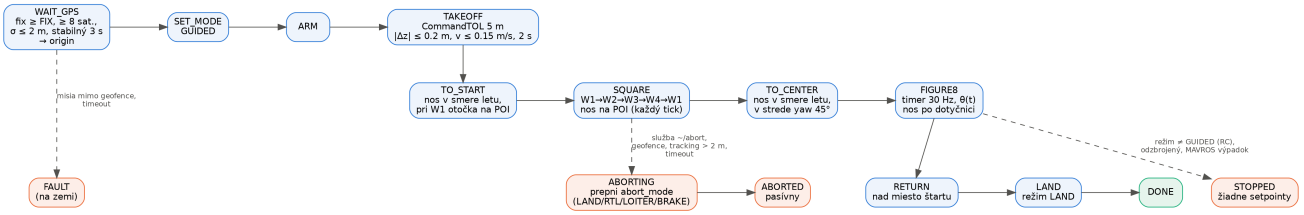
Clearance = distance from the drone centre to the nearest obstacle (walls, pillars, racks; not the floor), computed from the log for every airborne pose (`plot_a1_2.py`). The comparison flight with 0.30 m (`results/A1_2/safety_0.30_20261019/`) had only 0.20 m, which is why we use 0.45 m.





Commanded lines are blank during takeoff and landing on purpose: ArduCopter flies NAV_TAKEOFF / LAND itself and the executor sends no position setpoints then (a setpoint would cancel the takeoff). Route deviation is only defined while navigating.

A1.3 — Outdoor mission (GPS, compass, figure-8)



Origin from GPS and GPS ↔ metres

What: before takeoff we wait for a fix with at least 8 satellites and accuracy within 2 m, continuously for 3 s, and average all fixes in that time = the origin (launch point). Everything is in metres from it; there is no fixed GPS coordinate in the code. The conversion is a local tangent plane with WGS-84 radii at the origin (east = Δλ·(N + h0)·cos φ0, north = Δφ·(M + h0)), in both directions.

Why: the mean is more accurate than a single random fix. The mission spans less than 8 m; against the exact conversion through ECEF the error is 0.25 mm at 50 m and ~0.003 mm on our mission, hundreds of thousands of times below GPS noise. GPS is used **only for the origin**; the flight itself follows the smooth local position from ArduPilot.

Pros / cons: + works anywhere, no extra library. – A GPS jump during averaging shifts the whole mission; over kilometres the error would grow (500 m → 2.5 cm).

Angles and altitude

What	Convention
yaw in the code and in MAVROS	ENU: 0° = east, positive counter-clockwise
compass (`compass_hdg`)	0° = north, positive clockwise
ArduPilot	NED, converted by MAVROS

Conversion: **yaw = 90° – heading**, wrapped into ±180° (verified: compass 0° ↔ yaw 90°, in flight the two differed by 0.14° on average). "5 m" = 5 m above the launch point; we control the local altitude and also check `rel_alt` (difference ≤ 1 cm). Takeoff is confirmed when the drone is within 0.2 m of 5 m and nearly still for 2 s in a row (no fixed wait).

Square with the nose on the POI

What: corners W1 = NW → NE → SE → SW around the POI, side 5 m. A carrot with a trapezoidal speed profile (acceleration ≤ 0.5 m/s²) and its velocity as feed-forward. The nose turns 30 times a second to the bearing of the POI, computed from where the drone will be in 0.5 s (compensates the yaw lag). The square starts only when the nose is on the POI within 5°.

Why: the bearing to the POI changes all the time, fastest in the middle of a side. The smooth ramp removed sharp tilts at the corners; without feed-forward the drone lagged ~0.8 m, now ~0.1 m.

Pros / cons: + nose error at most 3.3°. – The 0.5 s look-ahead is an SITL estimate and needs tuning on the real drone.

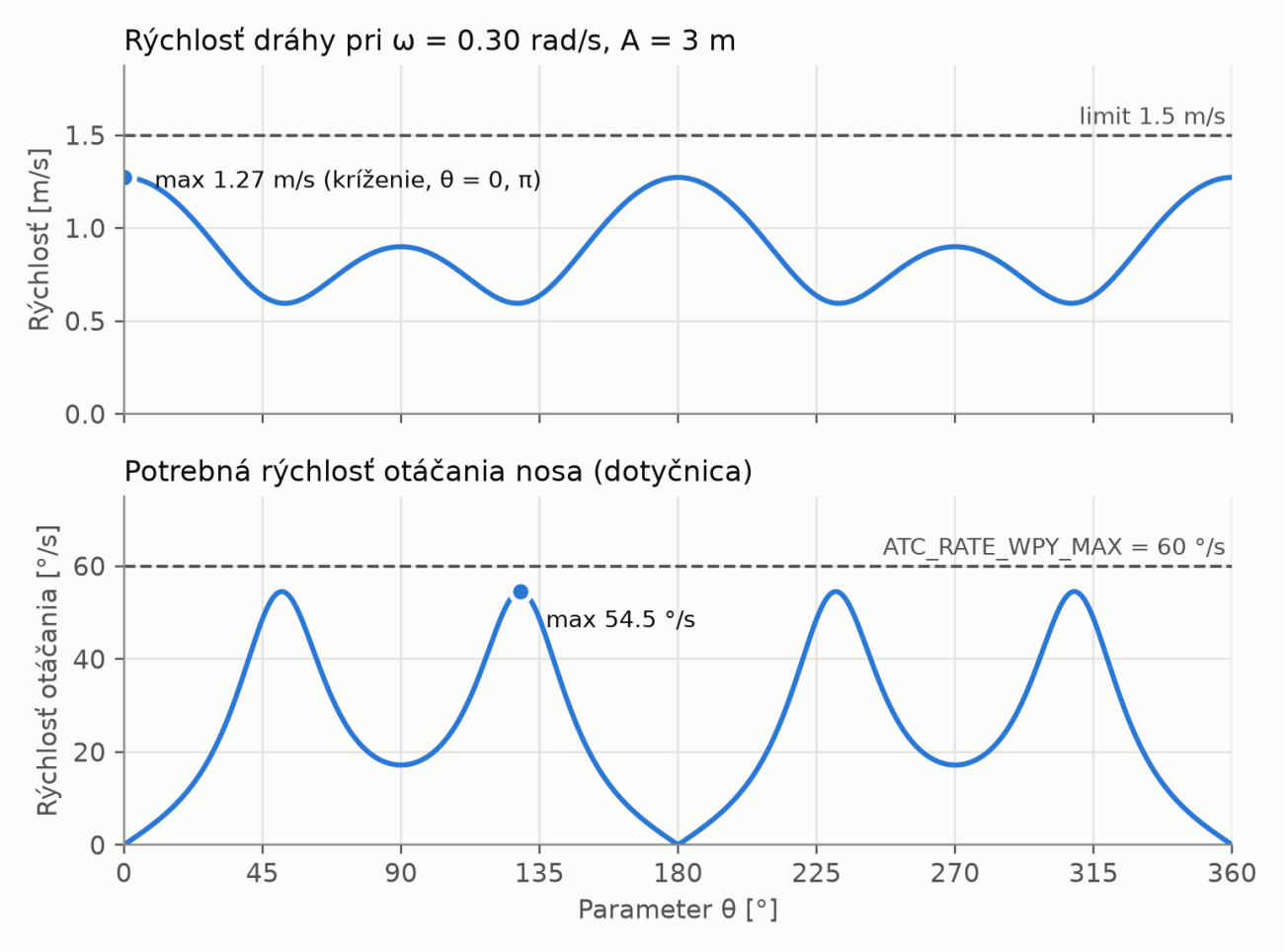
Figure-8 and tangent heading

What: no waypoints. A 30 Hz timer computes θ from time and the curve point `x = A sin θ`, `y = A sin θ cos θ` (A = 3 m, centred on the POI, 6 × 3 m). It sends position, velocity (the curve's derivative) and the tangent yaw `atan2(cos 2θ, cos θ)` with look-ahead and unwrapping. In the first and last 3 s the rate of θ changes smoothly.

Choice of ω: the top speed is `A·ω·√2` (at the crossing). A 1.5 m/s limit gives ω ≤ 0.35 rad/s; we chose **ω = 0.30 rad/s** → max 1.27 m/s, a lap in 21 s. At 0.35 the nose would have to turn at 64 °/s at the loop ends, above ArduPilot's 60 °/s limit (54.5 °/s at 0.30).

Geofence: the farthest points from the launch are the square corners (3.54 m); the limit is 4 m and 5.5 m altitude. The node checks this before flight and refuses to take off if the mission does not fit.

Pros / cons: + no spins or jerks, the angle corrects itself (not a yaw rate). – At the loop ends the nose lags up to ~24°, because the required turn rate is close to ArduPilot's limit.

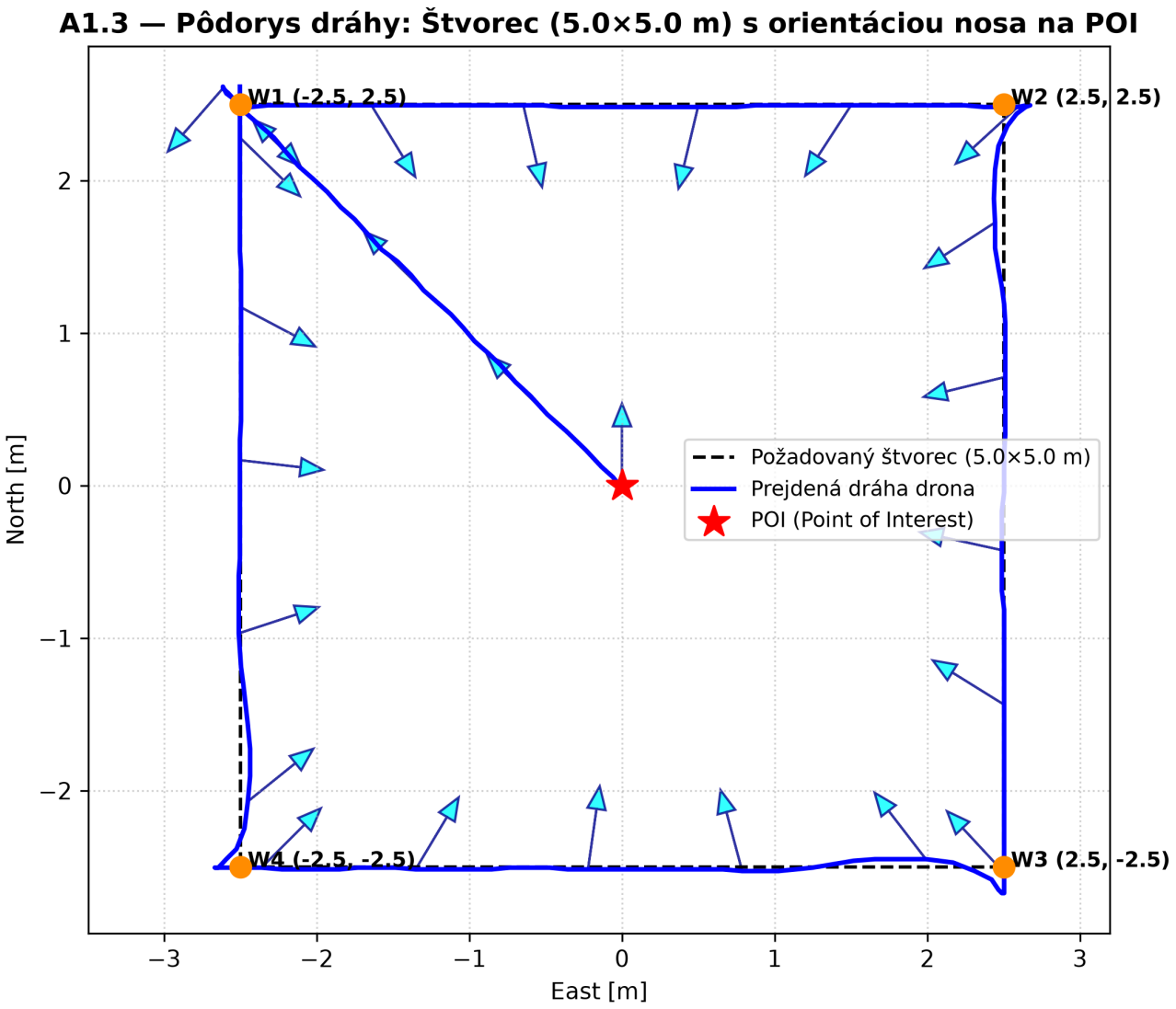


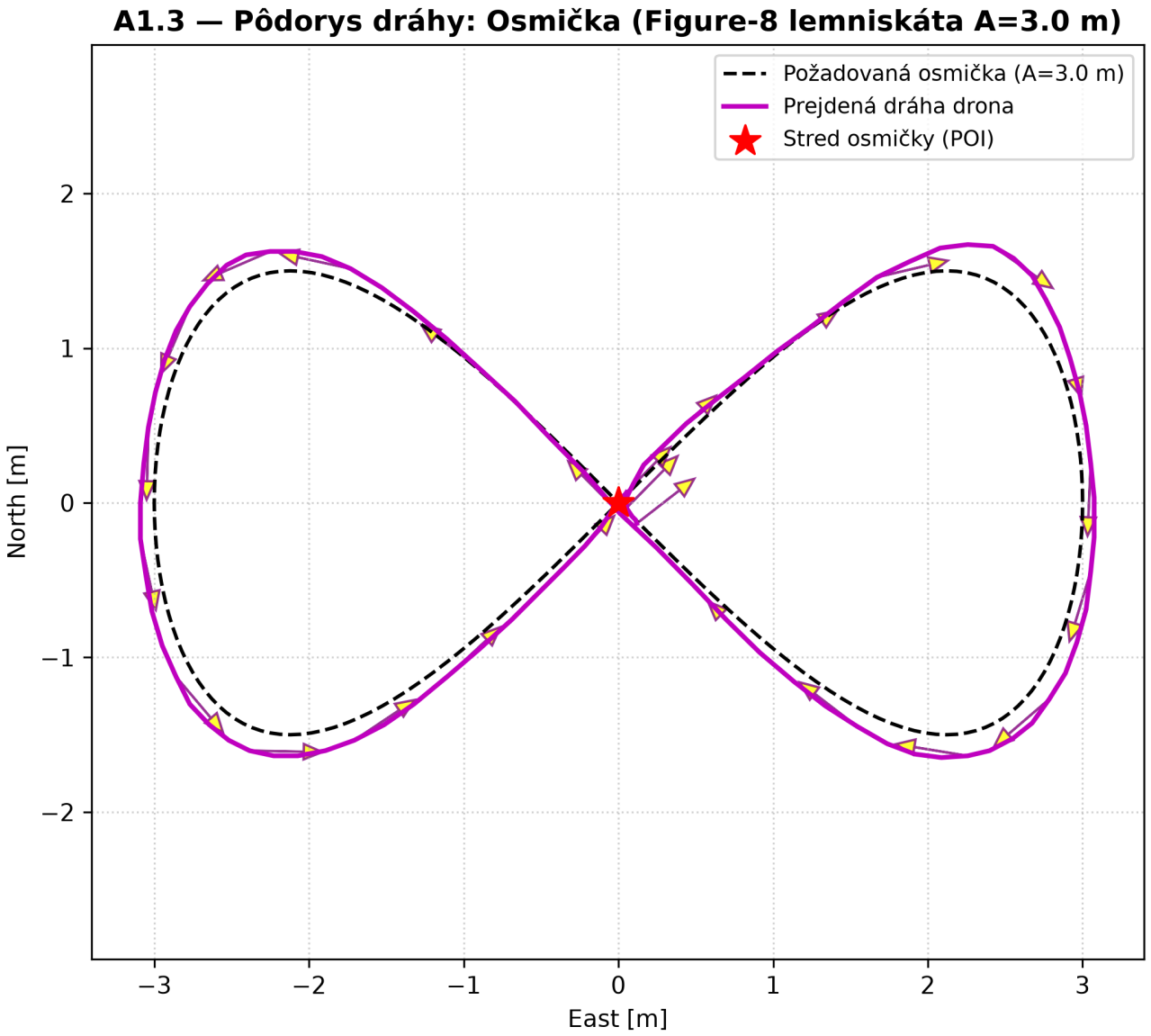
Landing, abort and safety

After the figure-8 the drone flies back over the launch point and switches to LAND. **Abort:** the `~/abort` service, leaving our own geofence or a tracking error above 2 m → LAND (or RTL/LOITER/BRAKE by parameter). When the pilot switches the mode, the drone disarms or MAVROS drops out, the node stops sending commands and switches nothing. The node runs in one thread (`rclcpp::spin`), so no mutex is needed.

A1.3 results (SITL, `results/A1_3/run_20261017_212049/`)

	Mean	Worst
square: nose vs. bearing to POI	2.0°	3.3°
square: deviation from path	0.09 m	0.31 m
figure-8: deviation from path	0.18 m	0.40 m
figure-8: nose vs. tangent	6.7°	23.7°
altitude	5.00 m	4.99–5.00 m
farthest from launch	—	3.71 m (limit 4 m)


A1.3 — Pôdorys dráhy: Štvorec (5.0×5.0 m) s orientáciou nosa na POI
W1 (-2.5, 2.5)
W2 (2.5, 2.5)
W3 (2.5, -2.5)
W4 (-2.5, -2.5)
Požadovaný štvorec (5.0×5.0 m)
Prejdená dráha drona
POI (Point of Interest)
North [m]
East [m]
2
1
0
−1
−2
−3
−2
−1
0
1
2
3



Coverage of the A1.4 requirements

Topic from the assignment	Where in this document
map representation, resolution	A1.1 › Map: 0.15 m voxel grid
inflation, radius value, racks	A1.1 › Racks, Obstacle inflation (0.45 m, verified in flight)
algorithm, heuristic, time, optimality	A1.1 › A*, A1.1 results
simplification, point counts, collisions	A1.1 › Path simplification, A1.1 results
position controller, hard/soft, "arrived", intermediate points	A1.2 › Carrot following, Hard and soft precision
heading policy, unwrapping	A1.2 › Carrot following and heading; A1.3 › Figure-8
geodetic conversion and its error	A1.3 › Origin from GPS
angle conventions (ENU, NED, compass)	A1.2 › Mission file and conventions; A1.3 › Angles and altitude
figure-8 parameters (ω, speed, geofence)	A1.3 › Figure-8
diagrams (data flow, ROS)	Overall solution diagram

Known limitations

- **A1.1 – holes in the roof of the cloud:** points are missing along the roof ridge; a goal above the roof (9, 6, 6.35) found a route outside. Mission points are inside the hangar; the holes could be closed with another box or a ceiling.
- **A1.1 – voxel rounding and sharp corners:** real clearance over the rack is 0.37 m instead of 0.45 m (a guaranteed 0.45 m would need ~0.53 m); no spline smoothing, so the drone briefly stops and turns at sharp corners.
- **A1.1 – NO_PATH** takes ~0.6 s, because A* has to search the whole reachable space.
- **A1.2 – small margin on the descent from the rack:** the drone sinks faster than it moves sideways (~0.15 m below the route); with 0.45 m still 4–5 cm above the drone radius.
- **A1.2 – after a soft waypoint** the next leg starts at the waypoint centre, so for the first metre the drone can be up to 0.4 m from the drawn route.
- **A1.2 – slow simulation:** on our PC Gazebo runs at 0.2–0.3× real time (start-up ~1.5 min, mission ~7 min); Gazebo must not be paused (SITL runs in lock-step).
- **A1.3 – nose at the figure-8 loop ends** lags up to ~24° (turn rate near ArduPilot's limit); the 0.5 s yaw look-ahead is an SITL estimate and needs tuning on the real drone.
- **A1.3 – a GPS jump during averaging** would shift the whole mission.